

Darrin Bright

Medical Image Analysis | Computer Vision | Deep Learning

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Research Interests

Medical Image Analysis | 3D Vision | Label-Efficient Learning

Education

Vellore Institute of Technology (VIT)

Integrated Master of Technology in Software Engineering, CGPA: 8.81/10

Vellore, India

Sep 2022 - May 2027

Relevant Coursework: Artificial Intelligence, Machine Learning, Statistics for Engineers, Soft Computing, Data Mining Techniques, Applied Discrete Mathematical Structures, Applications of Differential and Difference Equations.

Research Experience

Nanyang Technological University

Research Intern | Advisor: Dr. Si Yong Yeo

Singapore (Onsite)

Jul 2026 - Present

- Developing an end-to-end semi-supervised coronary vessel segmentation framework that treats segmentation as bidirectional correction, with complementary branches to recover missed vessels and remove false detections.
- Achieved preliminary Dice improvements of 2.34 pp on the XCA134 dataset and 1.63 pp on the XCAD dataset over the state-of-the-art baseline using only 10% labeled data.

Simon Fraser University

Research Intern | Advisor: Dr. Ghassan Hamarneh

Vancouver, Canada (Remote)

Dec 2025 - Present

- Developed the first deep learning method for direct detection of mitochondria-ER contact sites in 3D super-resolution microscopy (STED and SIM).
- Generated parametric 3D organelle geometries and rendered them through physics-based simulators to produce realistic training data, labeled with established contact-site criteria, overcoming the lack of ground truth in super-resolution microscopy.
- Validated on real microscopy against MCS-DETECT, correlative EM ground truth and known cell-line contact differences, showing strong spatial agreement with all three.

Indian Institute of Technology Kharagpur

Research Intern | Advisor: Dr. Debanjan Das

Kharagpur, India (Remote)

May 2026 - Jul 2026

- Developed GraphUNet, representing beat-level ECG/PPG morphology as graphs within a U-Net architecture for cuffless blood-pressure estimation, enabling explicit modeling of physiological relationships across heartbeats.
- Evaluated on VitalDB and PulseDB against published baselines, achieving 4.97 mmHg SBP MAE and 2.77 mmHg DBP MAE under subject-disjoint, calibrated evaluation, demonstrating competitive cuffless BP estimation performance.

Indian Institute of Technology Hyderabad

Research Intern | Advisor: Dr. C. Krishna Mohan

Hyderabad, India (Remote)

Jan 2026 - Jun 2026

- Built a federated continual learning framework integrating a global synthetic-data anchor with V-Information-guided selective re-distillation to preserve prior-task knowledge without sharing patient data or maintaining a conventional replay buffer.
- Evaluated across five MedMNIST modalities, improving average final accuracy by 32.12 pp and reducing average forgetting by 20.26 pp over FedCIL.

Vellore Institute of Technology

Undergraduate Researcher

Vellore, India (Onsite)

Jul 2025 - Jun 2026

SPHINX-Net: Histology-Aware Framework for Oral Cancer Detection

Advisor: Dr. Bharanidharan N.

- Developed SPHINX-Net, a lightweight hybrid CNN-Transformer framework for patch-level oral squamous cell carcinoma detection, combining histology-aware multi-view contrastive pretraining, spectral channel recalibration, and mini-batch prototype-graph refinement.
- Achieved 96.30% patch-level accuracy with a 1.75M-parameter model at 1,034 FPS.

PortionNet: Distilling 3D Geometric Knowledge for Food Nutrition Estimation

Advisor: Dr. Kanchan Keisham

- Developed PortionNet, a cross-modal knowledge distillation framework that transfers 3D geometric knowledge from point clouds to an RGB-only model through a lightweight RGB-to-Geometry Adapter, eliminating depth-sensor requirements at inference.
- Achieved SOTA on MetaFood3D, with 17.43% volume MAPE and 15.36% energy MAPE, reducing errors by 57.9% and 77.4% relative to the previous best; further achieved 12.17% energy MAPE on the SimpleFood45 dataset.

Indian Institute of Technology Jodhpur

Jodhpur, India (Onsite)

Research Intern | Advisor: Dr. Deepak Mishra

May 2025 - Jul 2025

- Developed a hierarchical vector quantization framework for skin lesion segmentation, pretraining on unlabeled ISIC 2018 dermoscopy images and using the learned discrete codes to regularize PH2 fine-tuning: 79.76% Dice, a consistent +1.38 pp over the VQ-free baseline across repeated runs.
- Ran a 13-configuration ablation of vector quantization inside Swin Transformer encoders on Pascal VOC 2012, achieving 74.61% Dice, +7.34 pp over the CNN baseline and +3.91 pp over the unquantized Swin.

Publications

PortionNet: Distilling 3D Geometric Knowledge for Food Nutrition Estimation

Journal of Computational Vision and Imaging Systems, Vol. 11, No. 1 (Proceedings of CVIS 2025) [Paper] [Code]

A Lightweight Histology-Aware Learning Framework with Spectral Channel Recalibration and
Prototype-Graph Refinement for Oral Squamous Cell Carcinoma Detection

Results in Engineering (Elsevier), Under Review

Technical Skills

Programming Tools	Python SQL Git LaTeX
ML & DL Tools	PyTorch TensorFlow OpenCV Matplotlib NumPy Keras MONAI
Medical Imaging	X-ray CT Histopathology Fluorescence Microscopy Dermoscopy

Work Experience

Astute

Vellore, India (Onsite)

Project Manager and Technical Lead, Intern

Oct 2024 - Apr 2025

- Architected SEO AgentFlow, a multi-agent AI framework with six specialized agents for keyword extraction, on/off-page optimization, technical SEO, and website analysis.
- Led development of AI-driven automation workflows integrating RAG-based content generation, web scraping, and automated content and advertisement generation.

GenAI/ML Engineer, Intern

Jul 2024 - Oct 2024

- Built a data-driven keyword research pipeline integrating SERP API, Google Trends, and DataForSEO with multi-dimensional ranking for competitive analysis.
- Developed automated advertisement generation using Stable Diffusion for poster design and Gemini for tagline generation and placement.

Leadership and Service

Chairperson, IEEE Robotics and Automation Society, VIT	Dec 2024 - Dec 2025
Volunteer, National Service Scheme	Dec 2023 - Dec 2024
Volunteer, Becoming I Foundation	Dec 2024 - Dec 2025

Honors and Awards

- Winner, Project 2039 Hackathon, Vellore Institute of Technology, 2024
- Winner, Alphaforge Ideathon, Vellore Institute of Technology, 2024
- Runner-up, Startup Demo Day, Vellore Institute of Technology, 2025
- Runner-up, Biomimicry Innovation Challenge, Vellore Institute of Technology, 2024
- Second Runner-up, HackWar Hackathon, Vellore Institute of Technology, 2025